

# 1.4 Network security

Where this unit sits, what it covers, and how far

## Where this unit sits

|                                    |                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification                      | GCSE Computer Science (9-1), specification code J277, from 2020                                                                                                                                                                              |
| Component                          | Component 01: Computer systems — 80 marks, 1 hour 30 minutes                                                                                                                                                                                 |
| What OCR says the component covers | the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software                                                                    |
| This unit                          | topic 1.4 Network security                                                                                                                                                                                                                   |
| Checked                            | 3 October 2026, at <a href="https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/">https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/</a> |

**How far this has been checked.** The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

**Before it:** 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols

**After it:** 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

## The lesson sequence and the questions it answers

| # | LESSON                                 | THE BIG QUESTION                                                              | ACTIVITIES | MARKS |
|---|----------------------------------------|-------------------------------------------------------------------------------|------------|-------|
| 1 | <b>Threat map</b><br>ns-101            | What forms of attack threaten a network, and what do attackers want?          | 11         | 23    |
| 2 | <b>Malware lab</b><br>ns-102           | How do the different types of malware spread, and what damage do they do?     | 12         | 19    |
| 3 | <b>Database break-in</b><br>ns-103     | How does an SQL injection attack work, and how can it be prevented?           | 12         | 19    |
| 4 | <b>Phishing inbox</b><br>ns-104        | How can you recognise a phishing message, and what should you do about it?    | 12         | 19    |
| 5 | <b>Social engineering HQ</b><br>ns-105 | What does a phishing attack set out to achieve?                               | 12         | 22    |
| 6 | <b>Cracking lab</b><br>ns-106          | How does a brute-force attack work?                                           | 13         | 18    |
| 7 | <b>Traffic flood</b><br>ns-107         | How does a denial-of-service attack work, and how can it be defended against? | 13         | 17    |

| #  | LESSON                                  | THE BIG QUESTION                                                                  | ACTIVITIES | MARKS |
|----|-----------------------------------------|-----------------------------------------------------------------------------------|------------|-------|
| 8  | <b>Packet sniffers</b><br>ns-108        | How is data captured in transit, and how can it be protected?                     | 12         | 19    |
| 9  | <b>Defence depot</b><br>ns-109          | Which protections stop which attacks, and why does no single one stop everything? | 12         | 19    |
| 10 | <b>Defence depot 2</b><br>ns-110        | How can we protect ourselves against hackers?                                     | 12         | 24    |
| 11 | <b>End-of-topic test</b><br>ns-111-test | assesses the whole unit                                                           | 0          | 0     |
| 12 | <b>Cyber defence</b><br>ns-bonus        | stretch work, no new content                                                      | 1          | 0     |

## Prior knowledge, vocabulary and skills

**Assumed before the unit:** the outcomes of 1.1, 1.2, 1.3

**Vocabulary introduced:** Malware, Social engineering, Virus, Ransomware, SQL injection, Input validation, Phishing, Blagging, Shouldering, Brute-force attack, Password, Denial of service, Botnet, Data interception, Encryption, Penetration testing, Firewall, Physical security

**What a pupil does, not only reads:** defence, match, mcq, multi, order, sort

## Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

**How this was worked out.** The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

| OUTCOME                                                                        | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|--------------------------------------------------------------------------------|--------|--------|-----------|----------|----------|
| understand the different forms of attack against computer systems<br>ns-101-o1 | ns-l01 | yes    | yes       | yes      | Full     |
| understand the threats of various forms of malware<br>ns-102-o1                | ns-l02 | yes    | yes       | yes      | Full     |
| understand how to identify and protect against malware<br>ns-102-o2            | ns-l02 | yes    | yes       | yes      | Full     |
| understand the concept of SQL injection<br>ns-103-o1                           | ns-l03 | yes    | yes       | yes      | Full     |
| understand how to protect against SQL injection<br>ns-103-o2                   | ns-l03 | yes    | yes       | yes      | Full     |
| understand what phishing is<br>ns-104-o1                                       | ns-l04 | yes    | yes       | yes      | Full     |
| understand how to identify and protect against phishing<br>ns-104-o2           | ns-l04 | yes    | yes       | yes      | Full     |
| understand what phishing is<br>ns-105-o1                                       | ns-l05 | yes    | yes       | yes      | Full     |

| OUTCOME                                                                                                                   | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|---------------------------------------------------------------------------------------------------------------------------|--------|--------|-----------|----------|----------|
| understand how to identify and protect against phishing<br><b>ns-105-o2</b>                                               | ns-l05 | yes    | yes       | yes      | Full     |
| understand the ways people are a weak point<br><b>ns-105-o3</b>                                                           | ns-l05 | yes    | yes       | yes      | Full     |
| understand what a brute-force attack is<br><b>ns-106-o1</b>                                                               | ns-l06 | yes    | yes       | yes      | Full     |
| understand how to identify and protect against brute-force attacks<br><b>ns-106-o2</b>                                    | ns-l06 | yes    | yes       | yes      | Full     |
| understand what a denial-of-service attack is<br><b>ns-107-o1</b>                                                         | ns-l07 | yes    | yes       | yes      | Full     |
| understand how to identify and protect against denial-of-service attacks<br><b>ns-107-o2</b>                              | ns-l07 | yes    | yes       | yes      | Full     |
| understand the threat that data interception and theft poses<br><b>ns-108-o1</b>                                          | ns-l08 | yes    | yes       | yes      | Full     |
| understand how to identify and protect against data interception<br><b>ns-108-o2</b>                                      | ns-l08 | yes    | yes       | yes      | Full     |
| understand the ways people are a weak point in system security<br><b>ns-109-o1</b>                                        | ns-l09 | yes    | yes       | yes      | Full     |
| know how penetration testing, anti-malware, firewalls and user access levels help against cyberattack<br><b>ns-109-o2</b> | ns-l09 | yes    | yes       | yes      | Full     |
| know how passwords, encryption and physical security help against cyberattack<br><b>ns-110-o1</b>                         | ns-l10 | yes    | yes       | yes      | Full     |
| be able to choose suitable protections for a given threat<br><b>ns-110-o2</b>                                             | ns-l10 | yes    | yes       | yes      | Full     |

20 full · 0 partial · 0 not covered, of 20 outcomes.

## Gaps, assumptions and overlap

None found by the trace.

### Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- ns-101-x3 A small business is hit by ransomware. Describe three costs it will face beyond the ransom itself.
- ns-102-x1 Name two types of malware and state what each one does.
- ns-102-x2 Explain why a worm can spread through a network faster than a virus.
- ns-102-x3 Describe three measures a school could take to reduce malware infections.
- ns-105-x2 Describe two measures that reduce the risk of social engineering.
- ns-106-x3 Explain why a long passphrase is harder to crack than a short complex password.
- ns-107-x3 Describe two measures that reduce the impact of a DDoS attack on a website.
- ns-108-x1 Describe how data can be intercepted as it travels across a network.
- ns-108-x2 Explain why encryption makes intercepted data useless to an attacker.

- ns-110-x3 A company has lost data to ransomware. Recommend three measures it should introduce, and justify each one.
- ns-111-test-x2 Malware: types, spread and damage
- ns-111-test-x7 Prevention methods and which threat each one stops

### **What could not be traced**

The wall text of ns-bonus is painted into the 360 image and held nowhere as text, so an outcome in that lesson can only be traced through its activities and its exam questions. Read the rows above for it with that in mind.

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